New Home Assignment

User Story Review

User Story - Contract Termination

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Background
Contract termination is a manual process in [redacted] Web platform. It requires a lot
of attention from contract managers (CM). In order to help CMs, our team needs to
automate this process.

Description
As a contract manager,
I should be able to specify the contract termination date on UI,
So the contract will automatically terminate on a defined date.

Acceptance criteria
* contract manager should be able to fill in the contract termination date via UI
* contract manager should be able to update the contract termination date via UI
* contract should automatically terminate on the defined date
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Task

1. Review the content of the user story and answer the following questions:
    - Is everything what needs to be implemented and tested clear?
    - if something is not clear, please describe missing parts and uncertainties.
2. Provide your version of acceptance criteria for the User Story based on provided by your assumptions
3. Provide a test approach with test cases for testing the feature
4. Provide an example map (a set of specific test data required for test execution) for one of the test cases created
5. Specify which tests have to be automated, on which level, and who should be responsible for implementation.

Please provide your solution in a word document.

Task

1. Review the content of the user story and answer the following questions:

- **Is everything what needs to be implemented and tested clear?**

As the user history is a first approach to a desired functionality and it is expected for the specific requirements to be documented at later stages, this is enough as a start point. I would only add an acceptance criteria regarding how the user is notified about the termination date.

- **if something is not clear, please describe missing parts and uncertainties.**

These are some of the points that can be clarified at later stages in terms of user expectations:

1. It is not clear whether a success message should appear for the user once the date is updated. Consequently, it is not clear if a failure message should appear in case the contract termination date cannot be set (for example in case of network error).

2. It is not clear what kind of notification should appear to the user after the termination date is reached.

2. Provide your version of acceptance criteria for the User Story based on provided by your Assumptions

Assumptions:

- It is not possible to update an expired contract.

- The functionality will not be optimized to be used on mobile devices.

- User can only have one session at the time, it is no possible to have race condition. In the same way, only one CM can access the same contract at a time and subsequently edit it.

- Auto save functionality for the termination date, no button to save changes is required.

- The component to be implemented for the user to fill in the contract termination date is a date picker.

- The date picker component guarantees by design that it is not possible to select dates with incomplete format.

- The system has access to an email service to send notifications.

- *System assumptions:*

    - The system has appropriate permissions and access controls to prevent unauthorized access to contract data.

    - The system has a database to store and manage contract data.

Acceptance Criteria:

- The contract manager can fill in the contract termination date using a date picker component on the UI.

- The contract manager can update the contract termination date via the same date picker component on the UI.

- Once the contract termination date is set, the contract should be marked with termination date set in the system.

- When the contract termination date is reached, the system should automatically terminate the contract, change its status to Terminated, and prevent any further modifications to the contract.

- The system should send an email notification to the contract manager and any other stakeholders when the contract is within 7 days of its termination date.

- If the contract termination date is updated, the system should automatically remove the termination date set mark in the system and cancel any pending email notifications.

- If the contract has already expired, the system should prevent the contract manager from setting or updating the termination date and display an error message indicating that the contract is no longer active.

- The system should keep a record of all contract terminations.

- After every valid change done by the CM in the date picker, the termination date is automatically saved.

3. **Provide a test approach with test cases for testing the feature**

Objective:
To ensure that the contract termination feature in the Deel Web platform is implemented correctly and functions as expected, enabling contract managers to specify the contract termination date via the UI and ensuring that contracts are automatically terminated on the defined date. Additionally, to ensure that the date picker component used in the feature works correctly across different browsers and platforms. The objective will be achieved by creating and executing a set of manual and automated tests, including integration tests, system tests, and acceptance tests, and by conducting code reviews and involving cross-functional team members in the testing process. For the manual testing, it is required as the functionality will be open to final users through the UI, and for the automation testing, as the contract termination is a Business-critical flow and with a high level of repeatability, this is a key procedure to improve and maintain quality for the feature.
The objective will be considered met when all tests have been executed successfully and all acceptance criteria have been met, indicating that the feature has been implemented correctly and meets the needs of the CM´s.
Testing level: For this component Integration Testing level will be considered among others test types described below.
Roles and responsibilities of each team member of the QA team:
QA specialist: test the feature to ensure that it meets the requirements and the acceptance criteria, develop, and execute test cases, perform manual testing, and report defects to the development team.
QA Automator/ DevOps: perform automated testing, ensure tests can be executed in a continuous integration/continuous deployment (CI/CD) pipeline.
Development team: As ISTQB states, the quality of the product is at the end responsibility of the whole team. Even though it fails out of the scope of this test approach, it is worth to mention the importance of unit tests and code reviews as tools used to identify and prevent defects early in the development process.

Types of Testing
Functional Testing: Verify that the contract termination feature works as expected from a final user standpoint.
Non-Functional Testing: Security Testing (to verify that the system prevents unauthorized access to modify the termination contract date), Recovery Testing (for example in case of network error during a termination date update).
UX/UI testing: UI testing (to Verify that the date picker component functions correctly on different browsers and platforms also viewport sizes, Responsive Design Testing (to check if the content scales to fit correctly),

Accessibility Testing (to ensure that the contract termination feature is usable by people with disabilities to some extent).
Integration Testing: Verify that the system sends email notifications to the contract manager and stakeholders when the contract is within 7 days of its termination date. Also, verify that the system records all contract terminations correctly in the database.

Test Cases per type:
Functional Testing
Verify that the contract manager can set the termination date of the contract using the date picker component.
Verify that the contract manager can update the termination date of the contract using the date picker component.
Verify that the contract is automatically terminated on the defined date.
Feature: Automated Termination date.
Precondition: CM is logged in with valid credentials and there is an existing open contract.
Scenario: Verify that the contract is automatically terminated on the defined date.
Given: User visits contracts section of Deel platform.
When: system date is <today´sDate>
Then: the contract should be automatically terminated and status should be updated to terminated.
Test data:

<today´sDate>	2023-03-13.

UX/UI testing
Verify that the date picker component is accessible for users with disabilities (baseline to be determined).
Verify that the date picker component works correctly on different browsers and platforms. (Baseline to be determined).
Feature: Automated Termination date.
Precondition: CM is logged in with valid credentials and there is an existing open contract.
Scenario: Verify that the date picker component works correctly on different browsers and platforms.
Given: User visits contracts section of Deel platform using <browser > and <platform>.
When: the CM clicks on the date picker component
Then: the date picker component should open and display correctly and the user should be able to select <validDate> and the selected date should be displayed correctly
Test data:

<browser>	Google Chrome	Mozilla Firefox	Apple Safari	Microsoft Edge	Opera
<platform>	Windows	MacOS	Linux		
<validDate>	2023-03-15.				

Verify that the UI updates as expected when the contract status changes.
Integration Testing
Verify that the system sends email notifications to the contract manager and other relevant parties when the contract is within 7 days of its termination date.
Feature: Automated Termination date.
Precondition: CM and other relevant parties have valid email addresses associated with their accounts and there is an there is an existing open contract marked with termination date set in the system.
Scenario: System sends notification emails to contract manager and relevant parties when the contract is within 7 days of its termination date.
Given: The contract termination date is 7 days from <today´sDate>.
When: the contract enters the 7-day notification window
Then: that the system sends email notifications to the contract manager and other relevant parties.
Test data:

<today´sDate>	2023-03-13.

Verify that the system records all contract terminations correctly in the database.

Non-Functional Testing:

Verify that the system prevents unauthorized access to the Contract termination feature.

Verify that appropriate permissions and access controls are for the Contract termination feature.

Verify that the system handles network errors correctly during a termination date update and provides appropriate error messages to the user.

Feature: Automated Termination date.

Precondition: CM is logged in with valid credentials and there is an existing open contract.

Scenario: System handles network errors and provides appropriate error messages to the user during a termination date update.

Given: User visits contracts section of Deel platform.

When the user attempts to update a contract termination date and a network error is simulated during the update process and the system is unable to complete the update

Then: The system should display an appropriate error <message> to the user and the system should log the error event and the system should automatically attempt to retry the update process

Test data:

<message>	"Unable to complete the update, Network error".

Negative Testing

Verify that the system displays an error message when the contract manager tries to set or update the termination date of an expired contract.

Feature: Automated Termination date.

Precondition: CM is logged in with valid credentials and there is an expired contract.

Scenario: System displays an error message when the contract manager tries to set or update the termination date of an expired contract.

Given: User visits contracts section of Deel platform.

When the user attempts to update a contract termination date of the expired contract.

Then: The system should display an appropriate error <message> to the user.

Test data:

<message>	"Can not update the termination date of expired contracts".

Verify that the system removes termination date set mark and cancels any pending email notifications when the contract termination date is updated.

4. Provide an example map (a set of specific test data required for test execution) for one of the test cases created

For the test case:

Verify that the contract manager can set the termination date of the contract using the date picker component.

Feature: Automated Termination date.

Precondition: CM is logged in with valid credentials and there is an existing open contract.

Scenario: Contract manager sets the termination date of the contract successfully.

Given: User visits contracts section of Deel platform.

When: CM selects the relevant contract

And CM selects the date picker component for the "Termination Date" picker and chooses the date <validDate>

Then: The system displays a confirmation message indicating that the contract termination date has been updated successfully to <validDate>.

Test data:

<validDate>	2023-03-15.

5. Specify which tests have to be automated, on which level, and who should be responsible for implementation.

Tests to be automated:

1. Functional Testing: as the contract termination is a Business-critical flow and with a high level of repeatability, this is a key procedure to improve and maintain quality for the feature.
2. Negative Testing: as it is a key feature for the business that the contracts are terminated in an accurate manner on all occasions.
3. UI testing: as it is necessary to make sure the date picker component functions correctly on different browsers and platforms. This will not include accessibility testing as it is planned to be run once with the use of accessibility tools such as Wave.
4. Integration Testing: as to test if the notifications are being sent by mail as expected and the database records the information accurately.

Due to the tests being developed with limited scope for the specific feature and the aim to test the interaction with external components like the database, they will be developed at integration level, as they will not involve various user scenarios at the same time, as it would be in a system testing.
In the same way, it is worth to mention the importance of unit tests for the date picker, as it will be on charge of the development team.
QA Automator/ DevOps roles will have the responsibility to implement the automated test cases and integrate them as part of a (CI/CD) pipeline, always thriving to implement maintainability and scalability to avoid significant changes in the future.